Catch Turtle All

In this project, you are required to develop and implement an interactive simulation in the Turtlesim environment using ROS2. Your main objective is to program a master turtle to "catch" other turtles that appear randomly within the environment. This project aims to test your proficiency with ROS2, focusing on topics, services, and action clients, and your ability to devise efficient algorithms for real-time robotic interaction in the Turtlesim simulation environment.

Task Requirements

System Requirements: The project must be implemented using ROS2 in conjunction with the Turtlesim simulation.

Turtle Generation: A new turtle should be spawned in a random location every 3 seconds.

Master Turtle Behavior: The master turtle should autonomously navigate towards and catch the nearest available turtle.

Turtle Chain Formation: Once a turtle is caught, it should automatically follow directly behind the master turtle, effectively joining a growing chain of turtles.

Simulation Loop: The nodes should run in an endless loop, and the entire simulation should be controllable via a launch file to start or stop the simulation.

Sub-tasks Breakdown

Sub-task A (20 Points): Implement the functionality to spawn a new turtle at a random location every 3 seconds.

Sub-task B (50 Points): Develop and integrate the autonomous path planning for the master turtle to catch the nearest turtle. This includes dynamic decision-making as new turtles spawn. NB: The turtle's actions only support left turn, right turn, and forward and backward movement, so the turtle's behaviours need to be planned according to the destination.

Sub-task C (30 Points): Program the behaviour for the caught turtles to follow the master turtle in a chain formation, maintaining close proximity to the turtle directly in front of them in the chain.

抓乌龟全部

在这个项目中，您需要开发并实现一个交互式模拟
使用 ROS2 的 Turtlesim 环境。你的主要目标是编程一个主乌龟去“抓住”环境中随机出现的其他乌龟。这个项目旨在测试你对 ROS2 的熟练程度，重点是主题、服务和动作客户端，以及你在 Turtlesim 仿真环境中为实时机器人交互设计高效算法的能力。

任务要求

系统要求：该项目必须使用 ROS2 并结合 Turtlesim 仿真来实现。

乌龟生成：每3秒应在随机位置生成一只新的乌龟。

掌握乌龟行为：主乌龟应自主导航并捕捉最近的可用乌龟。

乌龟链条形成：一旦捕获一只乌龟，它应自动紧跟在主乌龟后面，有效地加入不断增长的乌龟链条。

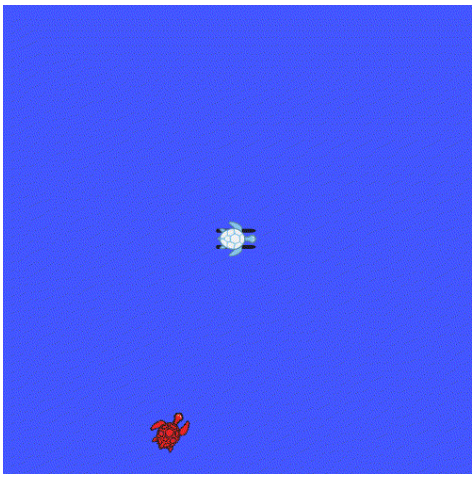
仿真循环：节点应在无限循环中运行，整个仿真应通过启动文件进行控制，以启动或停止仿真。

子任务拆解

子任务 A（20 分）：实现每 3 秒在随机位置生成一只新乌龟的功能。

子任务 B（50 分）：开发并集成主乌龟的自主路径规划，以追捕最近的乌龟。这包括在新乌龟生成时进行动态决策。注意：乌龟的动作仅支持向左转、向右转以及前进和后退，因此乌龟的行为需要根据目标地点进行规划。

子任务 C（30 分）：编写程序，使被抓住的乌龟以链式排列跟随主乌龟，保持与链中前方乌龟的接近距离。



Evaluation Criteria

Turtle Catching Efficiency: The master turtle must successfully identify and move towards the nearest turtle.

Chain Formation: The caught turtles must successfully attach and maintain formation directly behind the master turtle, forming a coherent chain.

Submission Requirements

Code Submission: Submit the project with all source code files used in the project.

Video Evidence: Provide a video recorded on a mobile phone or camera that captures the following elements (NB: video evidence cannot be edited and must be shot to the end.):

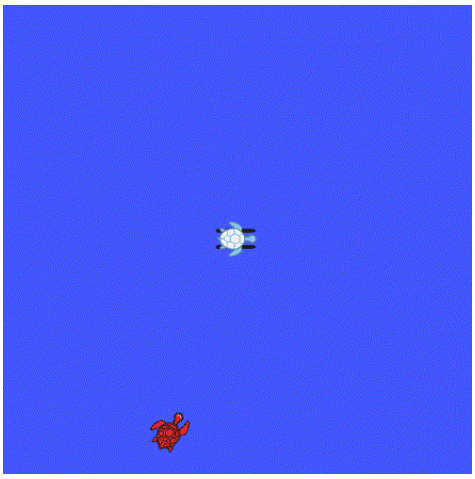
- **Team Members**: The video must include footage of all team members who contributed to the project.
- Robot Setup: Record the equipment running the Turtlesim simulation.
- Startup Process: Record the sequence of actions taken to start the robot.
- Main Functionalities: Showcase the primary functions of the robot as it operates. Showing how the robot fulfills each sub-task.

Project Report: Submit a detailed report that includes:

- An overview of your project and its objectives.
- A description of the robotic system and each of its components.
- An explanation of how each component contributes to the project objectives.

Group workload profile: Detail the contributions of each group member including:

- Name, student Id, and signature of each group member
- Specifying roles and responsibilities
- Individual contribution level i.e.:
  - 0% - I contributed the same as my team members
  - +X% I contributed more than my team members by a specific amount X% (see below)
  - -X% I contributed less than my team members by a specific amount X% (see below)
  - X% = 10%, 25%, 50%, 100%



评估标准

捕捉海龟的效率：主海龟必须成功识别并移动到最近的海龟。

链式排列：被抓住的乌龟必须成功地附着在主乌龟的正后方并保持排列，形成一个连贯的链条。

提交要求

代码提交：提交包含项目中使用的所有源代码文件的项目。

视频证据：提供一段通过手机或摄像机拍摄的视频，涵盖以下内容（注：视频证据不得编辑，必须完整拍摄至结束）：

团队成员：视频必须包含所有有贡献的团队成员的画面 d 项目。 l 机器人设置：记录运行 Turtlesim 仿真的设备。 l 启动过程：记录启动机器人的操作顺序。 l 主要功能：展示机器人运行时的主要功能。展示机器人如何完成每一个子任务。

项目报告：提交包含以下内容的详细报告：

l 您的项目及其目标概述。 l 机器人系统及其各个组件的描述。 l 解释每个组件如何为项目目标做出贡献。

小组工作负荷概况：详细说明每个组员的贡献，包括：

l 每个小组成员的姓名、学号和签名 l 指定角色和职责 l 个人贡献程度，即： n 0% - 我的贡献与组员相同 n +X% 我的贡献比组员多 X%（见下文） n -X% 我的贡献比组员少 X%（见下文） n X% = 10%，25%，50%，100%